9.6 Deployment considerations for AI web applications

Deploying AI web applications involves unique considerations beyond traditional web applications. The integration of LLMs introduces complexities such as cost management, data privacy, and latency that must be carefully addressed. Let's look at our options.

9.6.1 Deployment options

Deploying an AI web application to a production environment involves exposing it to the public internet at a designated URL address. But this process encompasses more than just the application itself. If your application provides data—such as chat histories or user-generated content—you will also need to configure databases for production use. Additionally, if you are utilizing external services like Clerk.js for authentication or Upstash for rate limiting, you need to configure these services appropriately for a production environment. Essentially, everything your application relies on must be reconfigured to ensure scalability and security.

There are several options available for hosting your application online.

Dedicated providers (Vercel)

Dedicated platforms like Vercel offer streamlined deployment workflows specifically designed for modern web applications, particularly those built with Next.js. Vercel provides a comprehensive suite of tools and infrastructure optimized for web applications, including AI web applications.

One of the reasons I frequently mention Vercel is its out-of-the-box experience that addresses many components of the production checklist we will discuss in the next section, such as content delivery network (CDN) integration, monitoring capabilities, and automated backups. For those just starting their journey with AI and Next.js, deploying on Vercel is highly recommended. It allows you to focus on development without getting bogged down in infrastructure management—at least until your application scales beyond what Vercel can accommodate. At that point, you may need to migrate some components or services outside of Vercel to meet your growing demands.

Containerization (Docker) and orchestration (Kubernetes)

For more advanced deployment needs, Docker and Kubernetes are powerful tools that facilitate the deployment of AI models while ensuring consistency across different environments. Docker allows you to package your applications into self-contained packages, which encapsulate all necessary dependencies and configurations. This approach guarantees that your application runs consistently regardless of where it is deployed.

Kubernetes, on the other hand, is basically a system that manages those containers at scale. It automates critical tasks such as load balancing, scaling based on demand, and rolling updates to ensure high availability and reliability.

Self-hosting

If you prefer not to use Vercel or have a preferred server-hosting provider, you can self-host your AI systems, which offers unmatched flexibility and customization. While